

SMP Workshop on Cavitation and Propeller Performances: The Experience of the University of Genova on the Potsdam Propeller Test Case

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ABSTRACT

Accurate and reliable propeller performances predictions are a fundamental aspect for any analysis and design of a modern propeller. Prediction of cavitation and cavity extension is another important task, being cavitation one of the crucial aspects that influences efficiency in addition to propagated noise and blade vibration and erosion. Accurate prediction of induced velocity field downward the propeller plane is, moreover, mandatory for the proper design of appendages and rudders. The Potsdam Propeller Test Case within the SMP Workshop on Cavitation and Propeller Performances represents an excellent chance to test and to validate the capabilities of the University of Genova, together with its numerical tools, in predicting propeller open water performances and cavitation.

Keywords

CFD, Panel Method, Cavitation, Propeller Performances.

1 INTRODUCTION

The design and the analysis of modern, loaded and highly skewed propellers is a primary aspect in the naval architecture field. The growing demand of heavily loaded, highly efficient propeller with request of very low noise and vibration levels onboard makes the design of marine propellers more and more conditioned by the analysis of inception and developed cavitation. Cavitation, in fact, is the more important inhibitor to the propulsion system. If not specifically scheduled, as in the case of supercavitating propellers, cavitation can generate a number of problems, like additional noise, vibrations and erosions as well as variations in the developed thrust and torque. Moreover, on modern propellers design, midchord bubble cavitation may also appears and face cavitation is very common, especially if the propeller operates behind a strong nonuniform wake or in inclined shaft conditions. Propeller performances, in addition, affect the efficient

design of rudders and appendages that have to operate, hopefully in cavity free conditions, inside the wake field behind the propeller itself. The prediction of the downstream velocity distribution produced by the propulsor, with special attention to the developed tip vortex, is, as a consequence, another crucial aspect for an accurate and useful analysis.

From a numerical point of view it is comprehensive the need of both accurate and efficient solvers, able to address the problem of performances and cavitation prediction in the preliminary design stage and in the more accurate final propeller optimization phase, during which features like tip vortex detachment or cavity bubble dynamics are taken into account. While in daily practice design boundary elements methods are usually applied because of their extreme computationally efficiency and their overall accuracy in both non cavitating and cavitating conditions for an acceptable range of advance coefficient around the design point, the increase in hardware resources leads to a shift towards viscous solvers for a more detailed propeller analysis. Major advantages of these approaches include the modeling accuracy, the amount of detailed information extracted from the simulations and the possibility of full scale analysis.

The Potsdam Propeller Test Case within the SMP Workshop on Cavitation and Propeller Performances represents an excellent chance to test and to validate the capabilities of the University of Genova, together with its numerical tools, in predicting propeller open water performances and cavitation. In particular two different approaches and three different solvers have been applied for the computation of open water propeller performances and cavitation prediction. The first approach is a potential based boundary element method developed at the University of Genova by Gaggero (2010), widely validated in steady and unsteady conditions

(Gaggero, 2008, 2010), including also cavitation. This approach will be adopted for the prediction only of the Potsdam Propeller open water curves and of the cavity extension. The approximate hypothesis on which the solver is based, in fact, does not make it suitable for a reliable prediction of the induced velocities and tip vortex, especially close to the mathematically “singular” trailing vortical wake, where a complex viscous interaction takes place. The second approach is based on the solutions of Reynolds Averaged Navier-Stokes equations through two different solvers. In the first case a commercial RANS solver, StarCCM+ v. 5.06 (CD-Adapco, 2010), has been applied for all the required computations: open water performances prediction, evaluation of downstream velocities and cavity extension. In the second case the open source CFD package OpenFOAM (OpenCFD, 2010) has been adopted for the computation, due to computational and time resources limitations, only of propeller open water characteristics.

2 NUMERICAL SOLVERS

2.1 Panel Method

Panel/boundary elements methods model the flowfield around a solid body by means of a scalar function, the perturbation potential $\phi(\mathbf{x})$, whose spatial derivatives represent the component of the perturbation velocity vector. Irrotationality, incompressibility and absence of viscosity are the hypothesis needed in order to write the more general continuity and momentum equations as a Laplace equation for the perturbation potential itself:

$$\nabla^2 \phi(\mathbf{x}) = 0 \quad (1)$$

For the more general problem of cavitating flow, Green’s third identity allows to solve the three dimensional differential problem as a simpler integral problem written for the surfaces that bound the domain. The solution is found as the intensity of a series of mathematical singularities (sources and dipoles) whose superposition models the inviscid cavitating flow on and around the body.

$$\begin{aligned} 2\pi\phi(\mathbf{x}_p) = & \int_{S_B+S_{CB}} \phi(\mathbf{x}_q) \frac{\partial}{\partial \mathbf{n}_q} \frac{1}{r_{pq}} dS \\ & - \int_{S_B+S_{CB}} \frac{\partial \phi(\mathbf{x}_q)}{\partial \mathbf{n}_q} \frac{1}{r_{pq}} dS \\ & + \int_{S_W} \Delta \phi(\mathbf{x}_q) \frac{\partial}{\partial \mathbf{n}_q} \frac{1}{r_{pq}} dS \end{aligned} \quad (2)$$

Neglecting the supercavitating case (computation is stopped when the cavity bubble reaches the blade trailing edge) and assuming that the cavity bubble thickness is small with respect to the profile chord, (Gaggero, 2009) singularities that model cavity bubble can be placed on the blade surface instead than on the real cavity surface, leading to an integral equation in which the subscript q corresponds to the variable point in the integration, $\mathbf{n}$ is the unit normal to the boundary surfaces and $\mathbf{r}_{pq}$ is the

distance between points p and q , S_B is the fully wetted surface, S_W is the wake surface and S_{CB} is the projected cavitating surface on the solid boundaries. This approach can be addressed as a partial nonlinear approach that takes into account the weakly nonlinearity of the boundary conditions (the dynamic boundary condition on the cavitating part of the blade and the closure condition at its trailing edge) without the need to collocate the singularities on the effective cavity surface. The set of required boundary conditions for the steady problem is:

- Kinematic boundary condition on the wetted solid boundaries:

$$\frac{\partial \phi(\mathbf{x}_p)}{\partial \mathbf{n}_p} = -V_{inflow}(\mathbf{x}_p) \cdot \mathbf{n}_p \quad (3)$$

- Kutta condition at blade trailing edge:

$$\Delta \phi_{TE}(\mathbf{x}_p) = \phi_{TE}^U(\mathbf{x}_p) - \phi_{TE}^L(\mathbf{x}_p) \quad (4)$$

- Dynamic boundary condition on the cavitating surfaces:

$$p = p_{vap} \text{ on } S_{CB} \quad (5)$$

- Kinematic boundary condition on the cavitating surfaces (where $\mathbf{n}$ is the normal vector and th is the local cavity bubble thickness):

$$\mathbf{n}(\mathbf{x}) - th(\mathbf{x}) = 0 \quad (6)$$

- Cavity closure condition at cavity bubble trailing edge.

Arbitrary detachment line, on the back and/or on the face sides of the blade can be found, iteratively, applying a criteria equivalent, in two dimensions, to the Villat-Brillouin cavity detachment condition, as in Mueller (1999). Starting from a detachment line obtained from the initial wetted solution (and identified as the line that separates zones with pressures higher than the vapor tension from zones subjected to pressure equal or lower pressures) or an imposed one (typically the leading edge), the detachment line is iteratively moved according to:

- If the cavity at that position has negative thickness, the detachment location is moved toward the trailing edge of the blade.
- If the pressure at a position upstream the actual detachment line is below vapor pressure, then the detachment location is moved toward the leading edge of the blade.

The numerical solution consists in an iterative scheme delegated to solve the nonlinearities connected with the Kutta, the dynamic and the kinematic boundary conditions on the unknown cavity surfaces until the cavity closure condition has been satisfied. Viscous forces, neglected by the potential approach, can be computed following two different approaches. In the first case, as

proposed by Hufford (1992) and Gaggero (2010), a thin boundary layer solver can be coupled, through transpiration velocities, to the inviscid solution, in order to obtain a local estimation of the frictional coefficient computed in accordance to the integral approach of Curle (1967) (for the laminar boundary layer) and Nash (1969) (for the turbulent boundary layer). This approach, though being applied successfully for the analysis of conventional propellers, poses some problems of convergence in very off design conditions and suffers from the tip influence on streamlines on which the boundary layer calculation is performed. As a consequence in the present work a local estimation of frictional coefficient has been carried out applying a standard frictional line. In particular, the VanOossanen formulation, based on local chord and thickness/chord ratio, has been employed:

$$C_D(r) = \frac{1}{2} a \left(1 + b \frac{t(r)}{c(r)} \right) Re_N^k(r) \quad (7)$$

A further force free condition for the trailing vortical wake can be employed, at least for the steady computations, requiring that each points move, from a Lagrange point of view, following:

$$\mathbf{x}(\hat{t}) = \mathbf{x}(\hat{t} - 1) + (\mathbf{V}_{inflow} + \nabla\phi) \cdot \widehat{\Delta t} \quad (8)$$

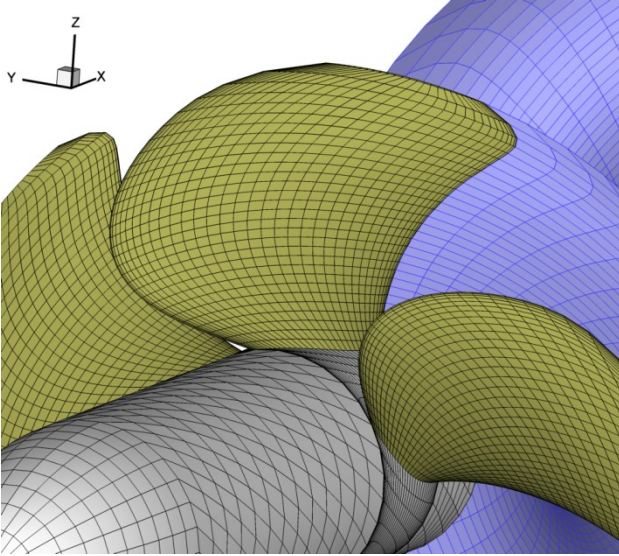


Figure 1: Panel Method propeller surface mesh.

In present work, however, only an approximate description of the trailing vortical wake dynamics has been adopted, and the trailing vortical wake has been modeled as a steady helicoidal surface whose pitch equals the local blade pitch.

The discretized surface mesh consists of 2500 hyperboloidal panels for each blade. The trailing vortical wake extends for five complete revolutions and it is discretized with a time equivalent angular step of 6° , as in figure 1.

2.2 RANS Solver – StarCCM+ and OpenFOAM

Analysis of open water propeller characteristics has been carried out through StarCCM+, a commercial finite volume RANS solver and by OpenFOAM, an open source CFD package. Continuity and momentum equation, for an incompressible flow, are expressed by:

$$\begin{cases} \nabla \cdot \mathbf{U} = 0 \\ \rho \dot{\mathbf{U}} = -\nabla p + \mu \nabla^2 \mathbf{U} + \nabla \cdot \mathbf{T}_{Re} + \mathbf{S}_M \end{cases} \quad (9)$$

in which $\mathbf{U}$ is the averaged velocity vector, p is the averaged pressure field, μ is the dynamic viscosity, $\mathbf{S}_M$ is the momentum sources vector and $\mathbf{T}_{Re}$ is the tensor of Reynold stresses. Cavitation is computed using a *volume of fluid* approach for the vapor fraction inside the domain, thus requiring the solution of a further transport equation for the percentage of vapor inside each cell. The bubble growth rate is based on the extended Rayleigh-Plesset equation (CD-Adapco, 2010).

Two turbulence closure equations models have been selected within StarCCM+. The *Realizable $k - \varepsilon$* and the *$k - \omega$* models have been adopted for the computation of open water and induced velocity propeller characteristics, in order to investigate the influence of the turbulence model on the non cavitating solution. For the computations in cavitating regime, only the more stable *Realizable $k - \varepsilon$* has been applied. OpenFOAM computations, instead, have been carried out adopting only the *$k - \omega$* closure equations.

The StarCCM+ computational domain, by means of the symmetries, is represented by an angular sector of amplitude $2\pi/Z$ around a single blade, discretized with unstructured meshes of polyhedral cells. OpenFOAM domain, due to the selected simulation model, instead is represented by a cylindrical region surrounding the entire propeller, discretized also in this case, using polyhedral elements. All the simulations, for both the solvers, have been carried out as steady, using the Moving Reference Frame approach. SIMPLE algorithm links pressure with velocity fields.

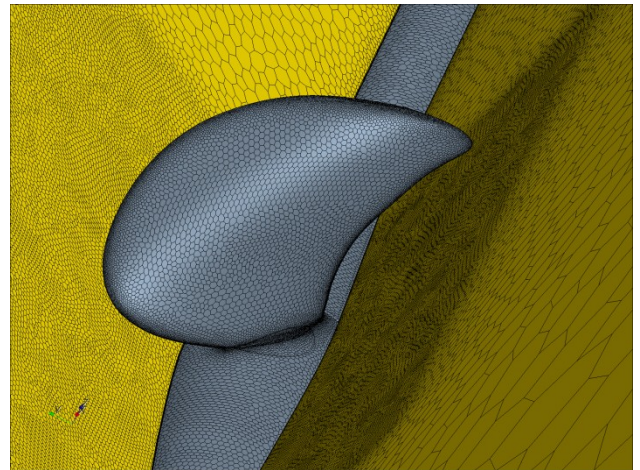


Figure 2: Typical unstructured Mesh for Open Water Computations. 1.25M/blade.

Depending on the peculiarities of the problem, different numbers of cells have been adopted. For the open water computations (case 2.1), the number of cells varies between 1.25M (StarCCM+, single blade, as in figure 2) to 1.4M (OpenFOAM, entire propeller). Due to some preliminary results, OpenFOAM computations have been performed on the sealed blade/hub configurations, neglecting the gap at the blade root (figure 3).

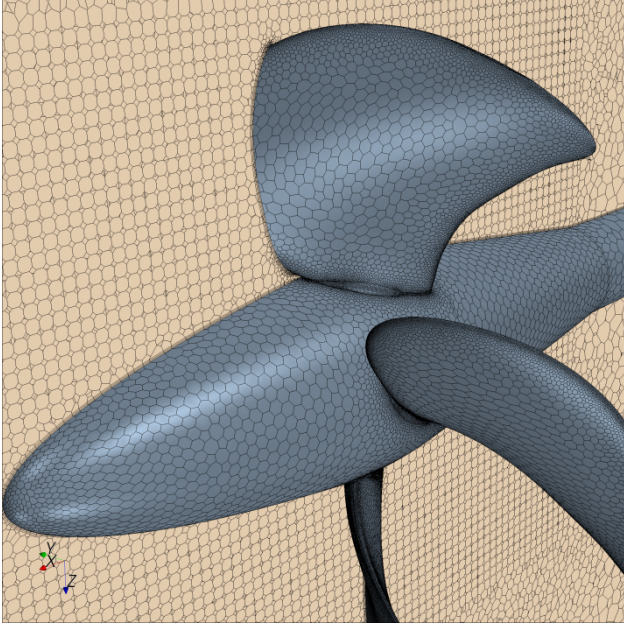


Figure 3: Typical unstructured Mesh for Open Water OpenFOAM Computations. 1.4M/entire propeller.

For the computation of induced velocities (case 2.2) a similar number of cells has been adopted. Moreover some refinements in the regions in which the velocity evaluation is required have been applied, increasing the number of cells up to 1.5M (StarCCM+). Finally for the cavitating flow (case 2.3) the solution has been computed on two different cells arrangements: a coarse grid, consisting in 1M elements for each blade, and a fine grid of about 2.4 million cells surrounding each blade. Also different meshing strategies, especially for what regard the prism layer thickness and discretization, have been selected, in order to reach satisfactory values for the y^+ parameter (generally lower than 30) and sufficient resolution for the cavitation model (between 3 and 10 on the blade).

3 RESULTS

3.1 Open Water – Case 2.1

Open water computations have been carried out with all the available computational tools. Prediction of thrust and torque are reported in figures 3 (thrust) and 4 (torque). It is worth to note that, for the thrust prediction, all the methodologies predict very close results. Only slight differences can be highlighted. StarCCM+ computations (despite the different $k - \varepsilon$ and $k - \omega$ turbulence models) are almost overlapped, and their slope is the same. Panel

method prediction, instead, is a bit overestimated with respect to the StarCCM+ computations, with some major differences at the lower value of advance coefficient. Due to the lack of leading edge separation forces prediction capabilities and the approximate viscous model, potential computations, as widely demonstrated (Gaggero, 2008, 2010), tend to be underestimated in working conditions (like lower advance coefficient conditions) in which viscous and separation effects are predominant. OpenFOAM predictions, unexpectedly for a viscous solver, are more close to the panel method results than to the StarCCM+ computations. Also for OpenFOAM it is possible to identify a slightly underprediction of thrust

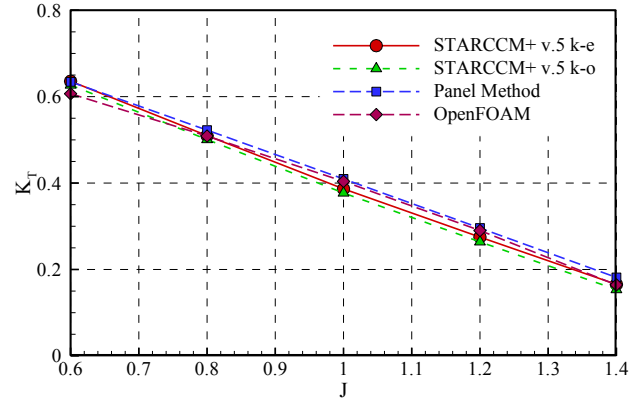


Figure 3: Open Water propeller characteristics. Comparison between Panel Method and RANS approaches (Thrust).

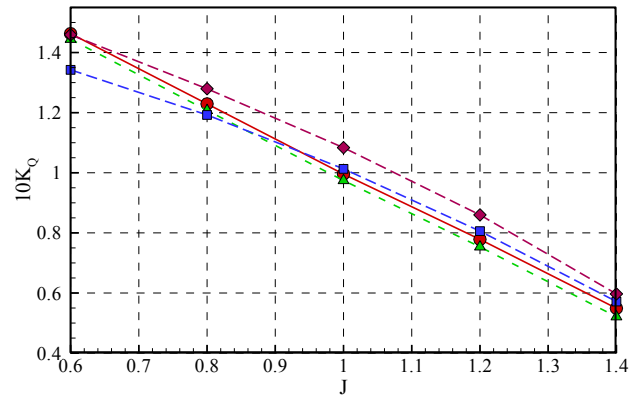


Figure 4: Open Water propeller characteristics. Comparison between Panel Method and RANS approaches (Torque).

coefficient for the lower advance value. Up to $J = 1.2$ OpenFOAM results are comparable with the Panel Method solution, while at higher value of advance, they are closer to StarCCM+. Also the slope of the curves (Panel Method vs. OpenFOAM) is similar and slightly different from the StarCCM+ ones.

For what regard the torque prediction, as presented in figure 4, it is possible to highlight some major differences between potential and viscous solutions. As for the thrust prediction, Panel Method solution is affected at lower values of advance coefficient by the limitations of the leading edge and viscous phenomena treatment. StarCCM+ results are very close, and also in this case the turbulence modelling seems not to have a significant

influence on the propeller hydrodynamic forces, with differences lower to 3% for almost all the advance range. OpenFOAM predictions, instead, have a different curve slope (more similar to the potential computations) together with very overpredicted values (with respect to the average value computed by StarCCM+ and as well by the panel method) of torque. A fine grid, even if a preliminary analysis with a different mesh arrangements did not show any significant improvement, could fill the gap with the other computational tools. In terms of pressure coefficient distribution, figures 5 and 6 compare the results from Panel Method (top), StarCCM+ (middle) and

potential solution, the Panel Method predicts the inversion of the angle of incidence, with negative values of pressure coefficient on the face side on the outermost strip of panels. Also for higher advance coefficient numerical solutions agree well. In this case, instead, it is the Panel Method pressure peak that is a bit underestimated at the blade leading edge (on the suction side, due to negative angle of attack) with respect to the viscous solutions. Also near the blade trailing edge there are some major differences, especially of the Panel Method and OpenFOAM solutions with respect to StarCCM+, for which the pressure increase on the face side (from $r/R = 0.7$ to the tip) is more emphasized.

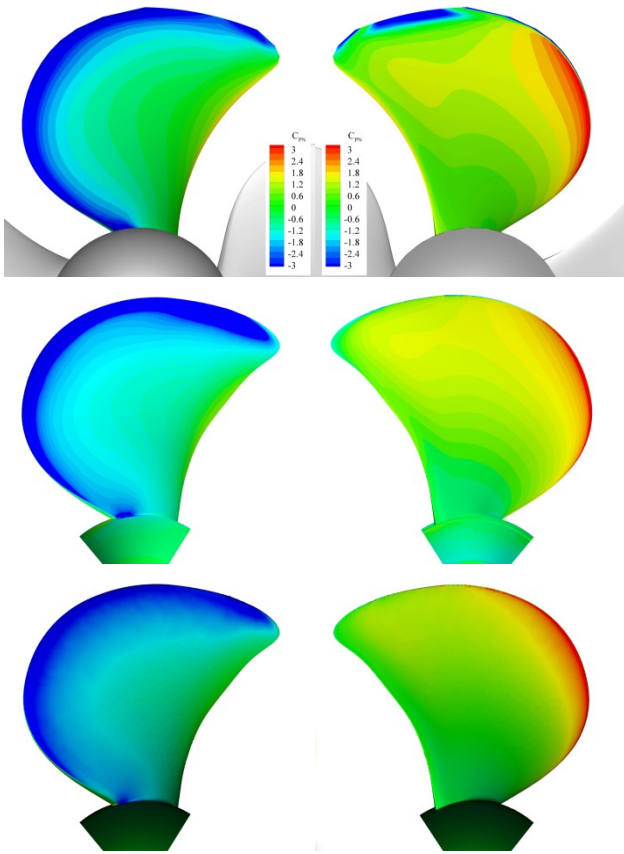


Figure 5: Pressure Coefficient distribution ($p/0.5\rho N^2 D^2$) at $J = 0.6$.

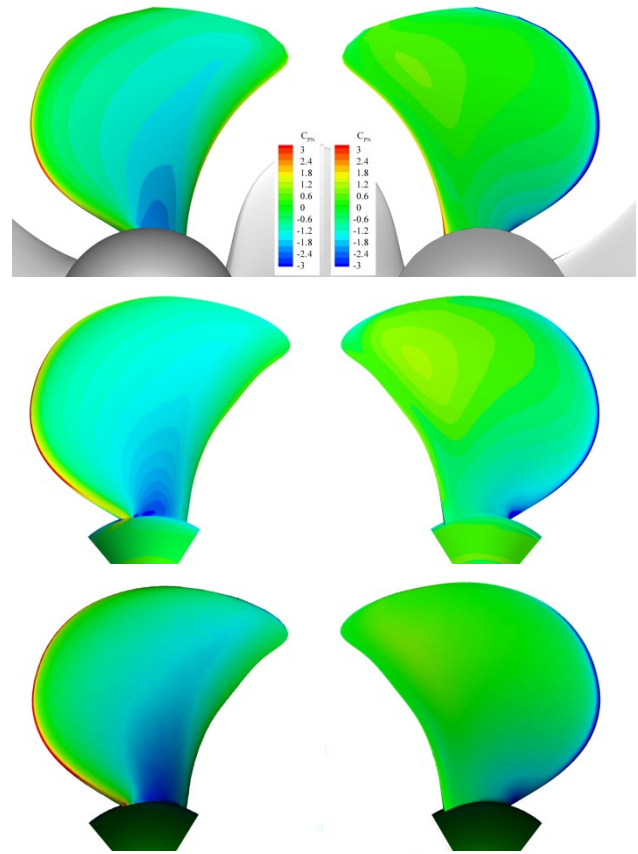


Figure 6: Pressure Coefficient distribution ($p/0.5\rho N^2 D^2$) at $J = 1.4$.

and OpenFOAM (bottom). As for the hydrodynamic coefficients, the agreement between computed values of pressure is good. At the lower advance coefficient ($J = 0.6$), where the load on the propeller is higher, the pressure peak on the face side predicted by the Panel Method is a bit overestimated (with respect to the StarCCM+ RANS solution). Also on the suction side Panel Method solution presents lower values of pressure, especially at the blade root where a clear difference in pressure distribution can be identified. In general RANS computations (StarCCM+ and OpenFOAM) are close each other, with some more evident differences with respect to the potential solution. At the blade tip, due to the lower accuracy of very skewed panels adopted for the

3.2 Velocity Evaluation – Case 2.2

Evaluation of downstream induced velocities has been performed only with StarCCM+. Panel Methods and, more in general potential theories (Gaggero, 2008), can be applied for the computations of the downstream velocity field. Their accuracy, however, due to the singular behavior of the trailing vortical wake if the induction point falls onto a trailing vortical panel, the neglecting of viscous effects and the lack of capabilities in modeling the tip vortex, can be considered sufficient only if an initial estimation of the mean downstream inflow characteristics is required.

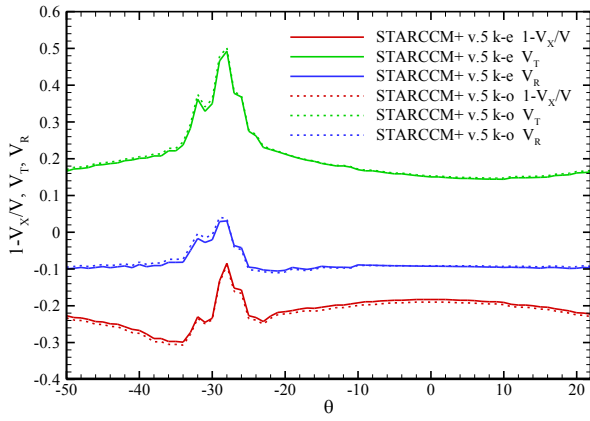


Figure 7: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 0.7$ and $x/D = 0.1$.

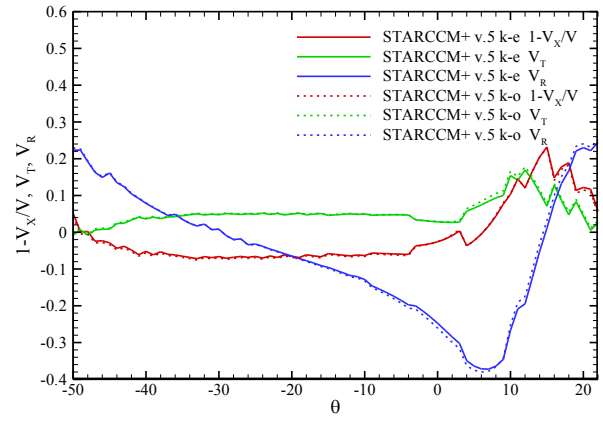


Figure 10: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 0.97$ and $x/D = 0.2$.

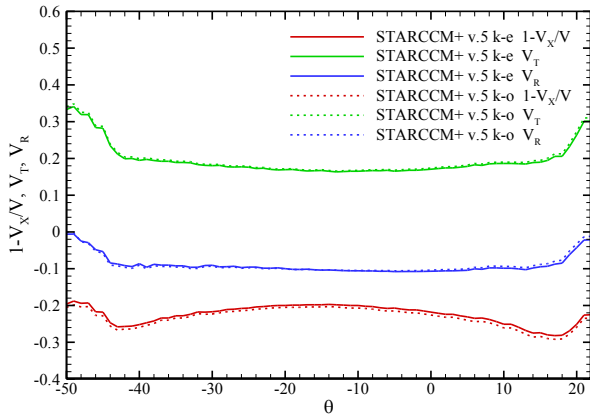


Figure 8: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 0.7$ and $x/D = 0.2$.

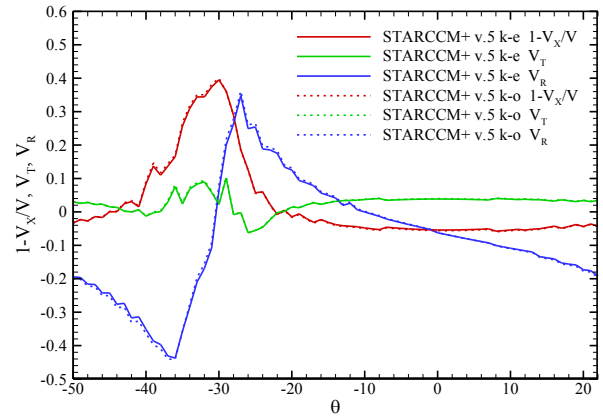


Figure 11: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 1.0$ and $x/D = 0.1$.

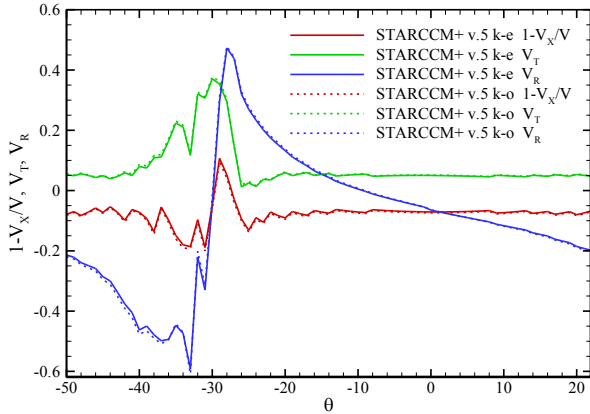


Figure 9: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 0.97$ and $x/D = 0.1$.

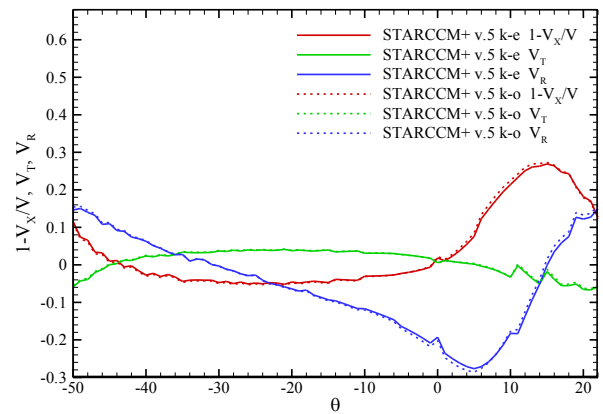


Figure 12: Axial, Tangential and Radial non dimensional velocity distributions at $r/R = 1.0$ and $x/D = 0.2$.

On the contrary, if the detailed solution of the tip vortex is the objective of the analysis, only RANS approach can provide reliable information. As suggested, all the computations have been carried out as steady and in accordance with thrust identity ($K_T = 0.250$) with a fixed rate of revolution equals to 23 *rps*. Both the *Realizable k-ε* and *k-ω* turbulence models have been applied. Non dimensional axial, tangential and radial velocities distributions downstream the propeller plane ($x/D = 0.1$ and 0.2) at $r/R = 0.7, 0.97$ and 1.0 are shown in figures from 7 to 12. Velocity distributions on the entire plane $0.1D$ and $0.2D$ aft the propeller origin are

reported in figure 13 to 16. As for the propeller performances prediction, the application of different turbulence models, as embedded in StarCCM+, does not change significantly the velocity distributions and the relative difference between the two computations is hardly distinguishable. The presence of the tip vortex is clearly identifiable, from the sudden variation of all the components of the velocity, close to the -30° position (at $x/D = 0.1$) and to $+10^\circ$ position (at $x/D = 0.2$). The non-smoothness of the distributions, especially close to the propeller plane, suggest the need of even more fine grid arrangements.

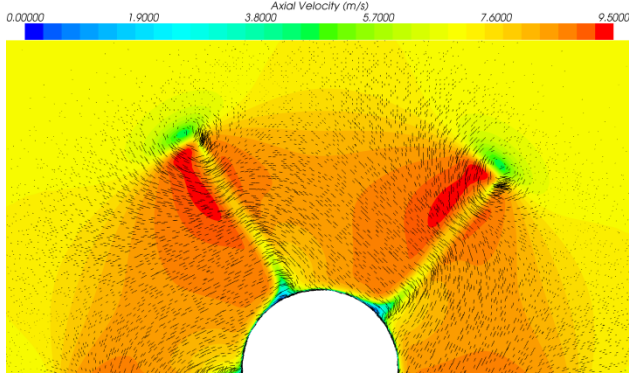


Figure 13: Velocity distribution at $x/D = 0.1$. $k - \epsilon$ turbulence model. From aft.

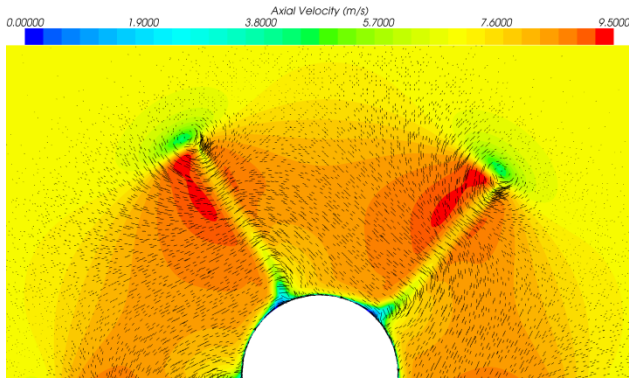


Figure 14: Velocity distribution at $x/D = 0.1$. $k - \omega$ turbulence model. From aft.

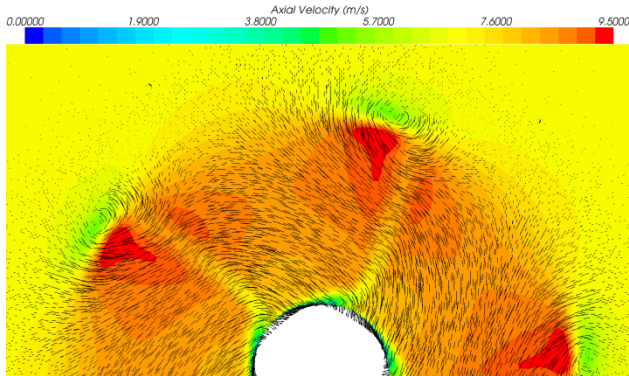


Figure 15: Velocity distribution at $x/D = 0.2$. $k - \epsilon$ turbulence model. From aft.

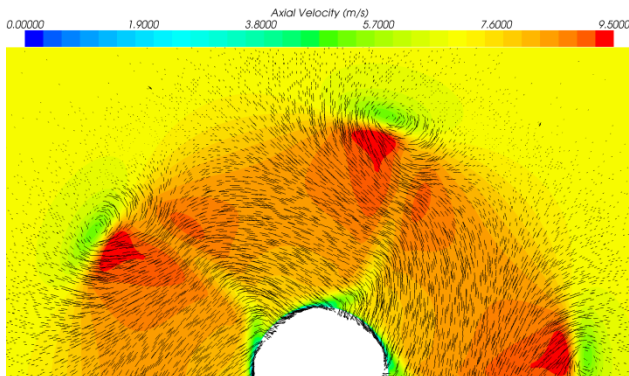


Figure 16: Velocity distribution at $x/D = 0.2$. $k - \omega$ turbulence model. From aft.

3.3 Cavitation – Case 2.3

Propeller performances and bubble extension evaluation in cavitating conditions has been carried out for three different cavitation indexes. Thrust identity has been selected in order to replicate the same tunnel measured flow conditions. As reported in the following tables, analyzed noncavitating thrust coefficients are 0.387, 0.245 and 0.167, with prescribed cavitation index (respectively 2.024, 1.424 and 2.000) and inflow velocity free to alter to match the thrust equivalence. As for the computations of induced velocities, all the calculations have been carried out in steady flow conditions, even if the experimental measures have been performed with the propeller inside a non-axial symmetric tunnel cross section. A cylindrical, same cross sectional area, section and the thrust identity hypothesis should, however, approximate accurately the real flow field. The initial coarse mesh (1M/blade) has been used to identify the regions subjected to cavitation. Refinements of the mesh (up to 2.4M/blade) have been applied locally on the light of the previous results and all the computations have been carried out with the finer discretization.

The analysis of the results for all the three conditions suggests some common consideration about the application of the selected flow solver for the non cavitating and the cavitating cases.

The predicted non cavitating pressure distributions by the Panel Method and by the RANS solver are in very well agreement. The only remarkable differences, as already highlighted in Gaggero (2008, 2010), can be identified at the blade leading and trailing edge. Absence of viscous effects and the application of the Kutta condition produces, for the Panel Method, higher values of pressure at the blade trailing edge, that slightly influence the pressure distribution also upstream on the profile. Near the leading edge difference between the two methodologies are less evident, especially for $r/R = 0.7$. If, instead, sections closer to the blade tip ($r/R = 0.9$ and $r/R = 0.95$) are considered, differences become (mainly for a cavitating analysis) non negligible. In this case the geometrical approximations of the panel description of the blade surface are responsible of these discrepancies: higher pressure peak at leading edge (as in figure 31, 32, 33 and 41, 42, 43) and local inversion of the incidence angle (figure 5). If cylindrical sections are adopted for the discretization of propeller blade, at blade tip hyperboloidal panels, especially if tip chord is zero, have very high aspect ratio and skewness factors, whose effect is to reduce the accuracy in the calculation of influence coefficients and, in turn, in the prediction of potential pressure distribution.

Also in the case of cavitating flow the comparison between Panel Method and RANS computations shows a general good agreement. Except some differences in term of cavity extension, both the solvers in each of the three considered conditions predict similar cavity pattern. In

Table 1: Case 2.3.1 propeller performances

	Tunnel	RANS	Panel
J	1.019	1.011	1.009
σ_N	2.024	2.024	2.024
K_T (non cav.)	0.3870	0.3870	0.3870
$10K_Q$ (non cav.)		0.9876	0.9679
K_T (cav.)		0.3782	0.3922
$10K_Q$ (cav.)		0.9724	1.0023

kind of phenomena higher, both the solvers detect a region with pressure below the vapour tension with a cavity bubble which extension is similarly computed in the radial and in the chordwise direction. Also at the tip the Panel Method is affected by some limitations regarding the capability of capturing tip vortex cavitation that, instead, can be deal within the RANS approach. However, if cavity bubble thickness at blade tip (for instance that of figure 18) can be considered a measure of the risk of tip vortex cavitation, also Panel Method can give an estimation of this phenomena.

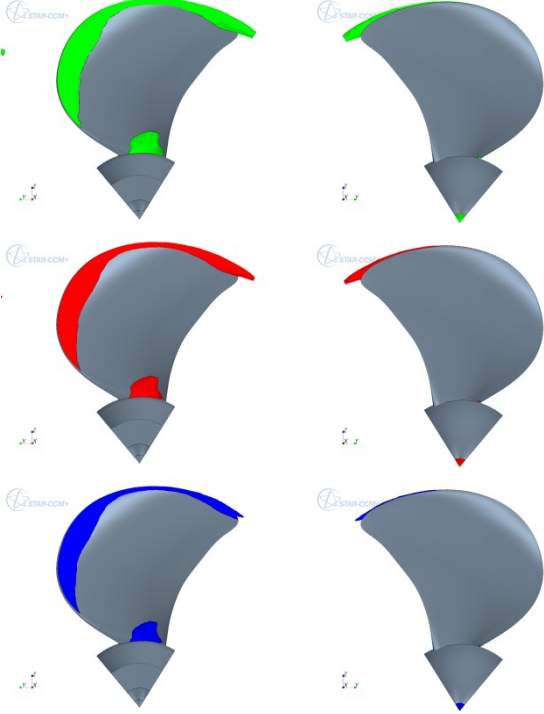


Figure 17: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.019$, $\sigma_N = 2.024$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

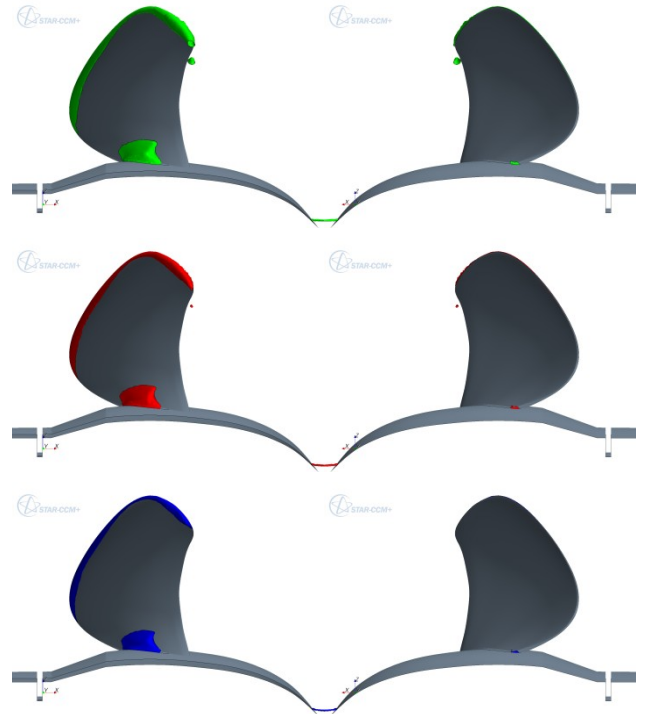


Figure 19: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.019$, $\sigma_N = 2.024$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

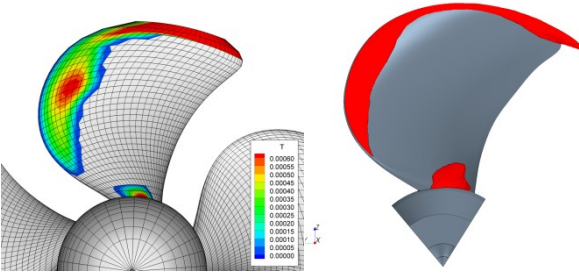


Figure 18: Cavitation on the suction side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.019$, $\sigma_N = 2.024$.

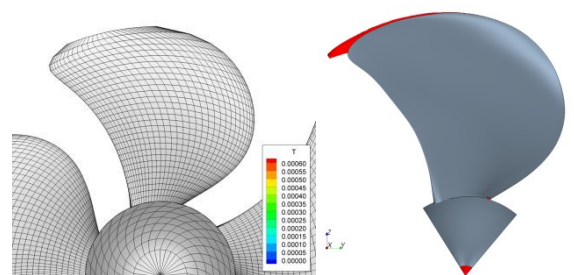
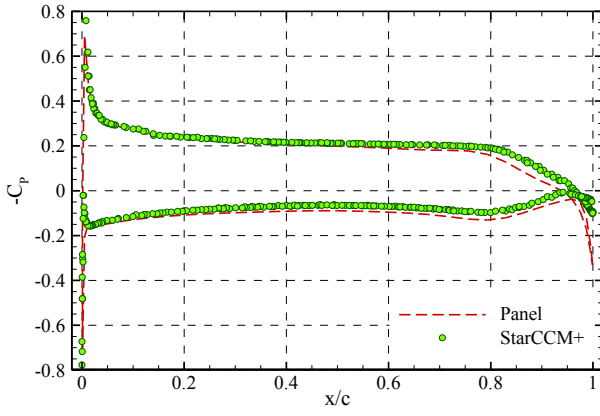


Figure 20: Cavitation on the pressure side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.019$, $\sigma_N = 2.024$.

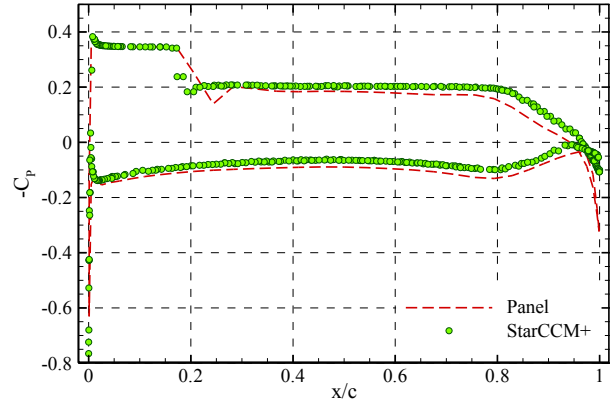
figures 18 and 20 the cavity extension computed by the Panel Method for case 2.3.1 is compared with the 50% isosurface of vapor fraction predicted by StarCCM+.

Also if bubble cavitation is not specifically addressed by the Panel Method and its presence is treated with the hypothesis of sheet cavitation, near the blade root, where thickness is greater and, consequently, the risk of this

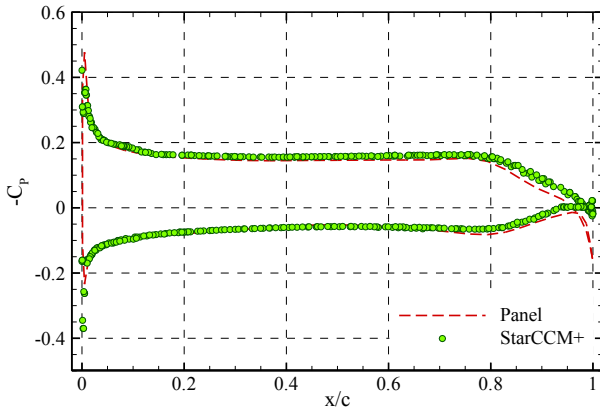
In terms of leading edge back sheet cavitation, instead, computed cavity extension is almost equivalent, with a slightly overestimation of the Panel Method solution in chord direction. The first cavitating section, in radial direction, instead, is pretty the same.



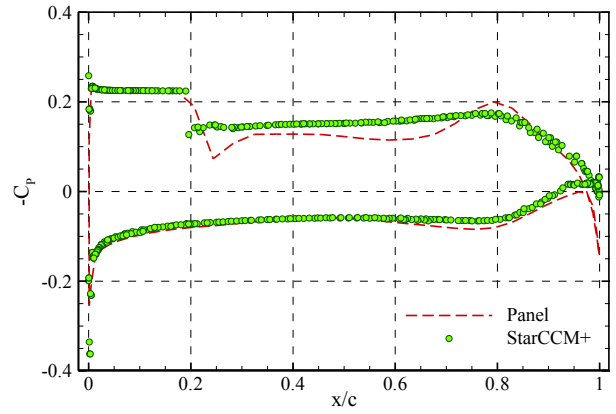
**Figure 21: Non cavitating pressure distribution. $r/R = 0.7$
 $J_{\text{tunnel}} = 1.019$.**



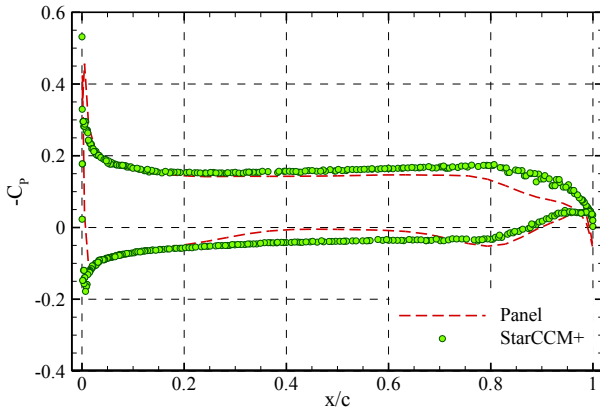
**Figure 24: Cavitating pressure distribution. $r/R = 0.7$
 $J_{\text{tunnel}} = 1.019$, $\sigma_N = 2.024$.**



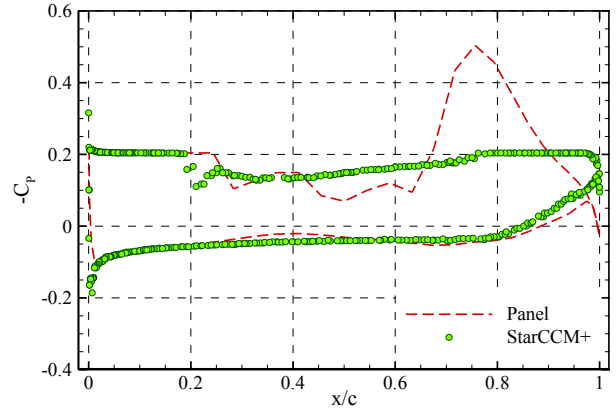
**Figure 22: Non cavitating pressure distribution. $r/R = 0.9$
 $J_{\text{tunnel}} = 1.019$.**



**Figure 25: Cavitating pressure distribution. $r/R = 0.9$
 $J_{\text{tunnel}} = 1.019$, $\sigma_N = 2.024$.**



**Figure 23: Non cavitating pressure distribution. $r/R = 0.95$
 $J_{\text{tunnel}} = 1.019$.**



**Figure 26: Cavitating pressure distribution. $r/R = 0.95$
 $J_{\text{tunnel}} = 1.019$, $\sigma_N = 2.024$.**

The agreement between cavity extensions is confirmed by the analysis of pressure distributions. For both the solvers, at $r/R = 0.7$ and $r/R = 0.9$, the constant vapor pressure region develops for about the 20% of the local blade chord and only the pressure recovery at the bubble trailing edge is overpredicted by the Panel Method. Close to the tip RANS solution is characterized by two different cavitating regions: the first, as usual, from the blade

leading edge and the second, close to the trailing edge, detaching aft midchord.

As Panel Method has been developed, it lacks in the capability of computing simultaneous (on the same radial section) leading edge and midchord cavity bubbles. As a consequence, also if leading edge bubble (and pressure distribution) is well in agreement with the viscous

Table 2: Case 2.3.2 propeller performances

	Tunnel	RANS	Panel
J	1.269	1.256	1.278
σ_N	1.424	1.424	1.424
K_T (non cav.)	0.2450	0.2450	0.2450
$10K_Q$ (non cav.)		0.7014	0.7067
K_T (cav.)		0.2035	0.2369
$10K_Q$ (cav.)		0.6365	0.7047

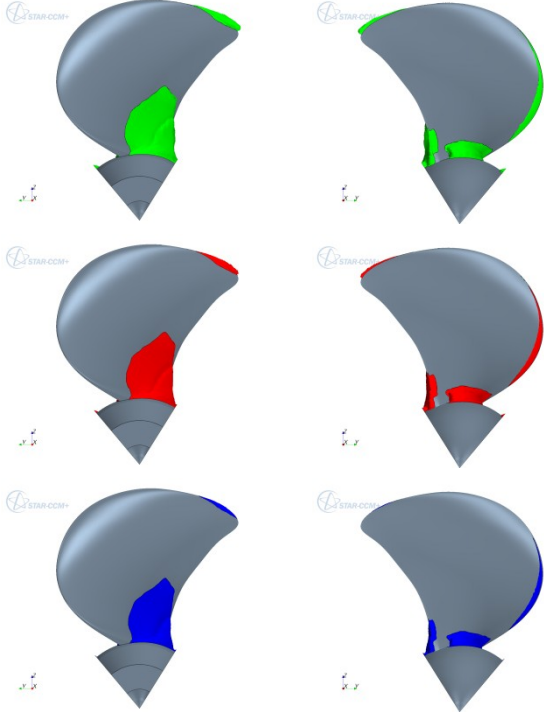


Figure 27: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.269$, $\sigma_N = 1.424$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

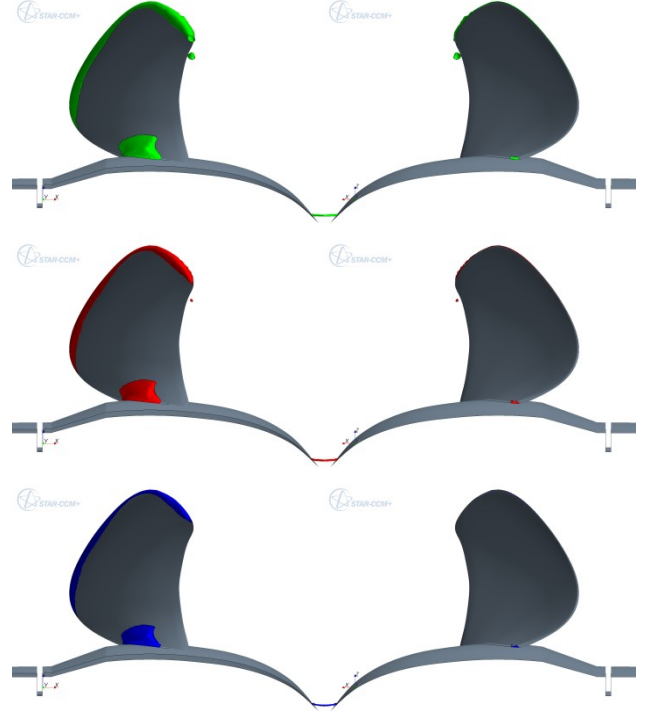


Figure 29: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.019$, $\sigma_N = 2.024$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

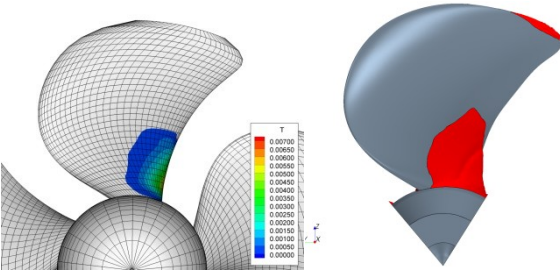


Figure 28: Cavitation on the pressure side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.269$, $\sigma_N = 1.424$.

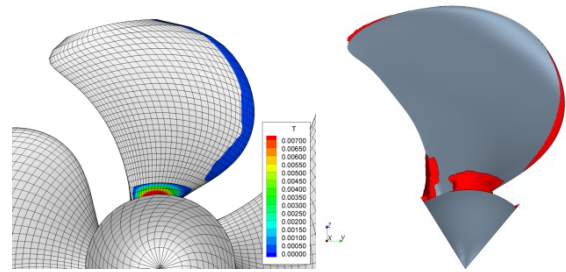


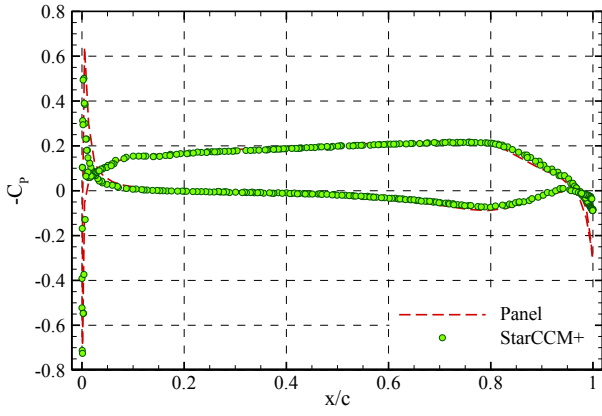
Figure 30: Cavitation on the suction side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.269$, $\sigma_N = 1.424$.

solution, at midchord the developed potential code could not include in the computations the presence of a newly detached bubble even if the resulting pressure distribution is characterized by values of pressure lower than the vapour tension (figure 26).

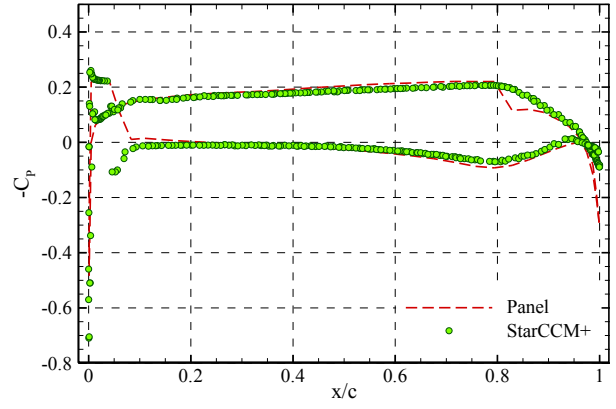
Case 2.3.2, instead, is numerically characterized by simultaneous back midchord cavitation and a thinner face

sheet cavity bubble from blade leading edge. Also for this inflow condition, results from the two approaches, as presented in the comparison of figures 28 and 30 (cavity extension, 50% vapour fraction) and of figures 31 to 36 (pressure distributions), indirectly validate the numerical computations. Both the solvers, in fact, although being based on very different theoretical basis, predict almost the same cavity features, even on the back than on the face side. A larger midchord cavitating region on the suction side extends up to the blade trailing edge from the blade root to $r/R = 0.65$.

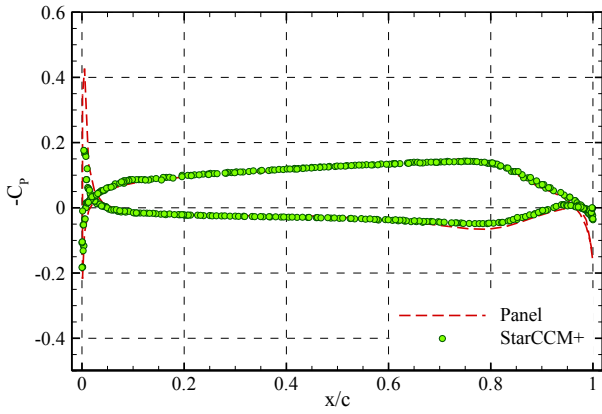
Also on the face side midchord root bubble is calculated by the two methodologies, with very similar extension and detachment line. Sheet cavitation on the pressure side is also predicted. As for case 2.3.1, Panel Method computations are a bit overestimated with respect to RANS results, especially for what regards the radial extension of the cavity bubble, that in the Panel Method



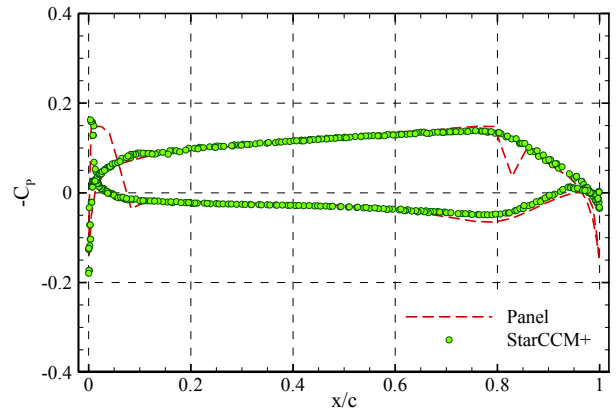
**Figure 31: Non cavitating pressure distribution. $r/R = 0.7$
 $J_{tunnel} = 1.269$.**



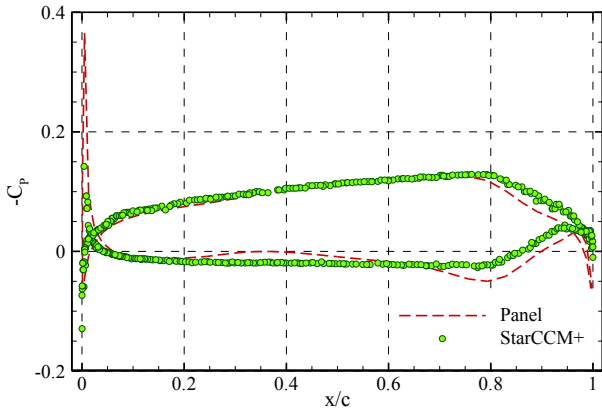
**Figure 34: Cavitating pressure distribution. $r/R = 0.7$
 $J_{tunnel} = 1.269$, $\sigma_N = 1.424$.**



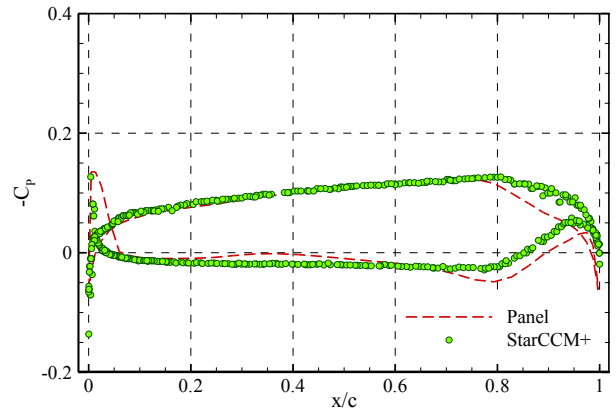
**Figure 32: Non cavitating pressure distribution. $r/R = 0.9$
 $J_{tunnel} = 1.269$.**



**Figure 35: Cavitating pressure distribution. $r/R = 0.9$
 $J_{tunnel} = 1.269$, $\sigma_N = 1.424$.**



**Figure 33: Non cavitating pressure distribution. $r/R = 0.95$
 $J_{tunnel} = 1.269$.**



**Figure 36: Cavitating pressure distribution. $r/R = 0.95$
 $J_{tunnel} = 1.269$, $\sigma_N = 1.424$.**

reaches sections close to the blade tip, while in the viscous computations it is limited to about $r/R = 0.8$.

The analysis of cavitating pressure distributions highlights, once more, the sufficient level of accuracy provided by the potential approach. As in the non cavitating case, pressure distributions are almost overlapped, with noticeable differences only at the blade

leading (higher pressure peak) and trailing (Kutta condition) edge. The lower values of pressure calculated by the Panel Method, especially close to the blade tip, are responsible of the higher amount of sheet cavitation predicted by this approach.

Table 3: Case 2.3.3 propeller performances

	Tunnel	RANS	Panel
J	1.408	1.387	1.422
σ_N	2.000	2.000	2.000
K_T (non cav.)	0.1670	0.167	0.167
$10K_Q$ (non cav.)		0.5396	0.5412
K_T (cav.)		0.1306	0.1378
$10K_Q$ (cav.)		0.4826	0.4939

distributions show a strong overprediction of the low pressure peak with respect to the RANS computation. As a consequence the risk of cavitation increases and the Potential Panel Method cavity prediction results in a slightly overestimation of the cavity extension. The analysis of the pressure distribution shows, moreover, the presence (on the Panel Method computations) of midchord cavitation not only at the blade root but also close to the tip ($r/R = 0.7 - 0.9$). Near 80% of chord (from profile leading edge) it is possible to note a sudden increase of pressure, typical of the cavity trailing edge

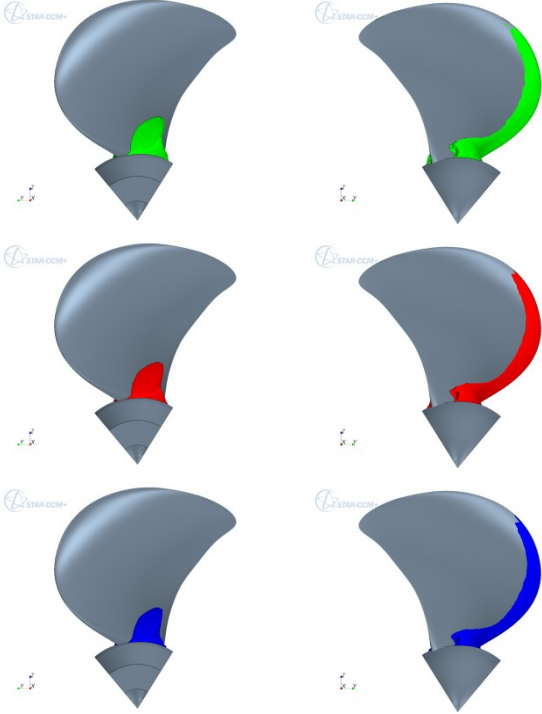


Figure 37: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.408$, $\sigma_N = 2.00$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

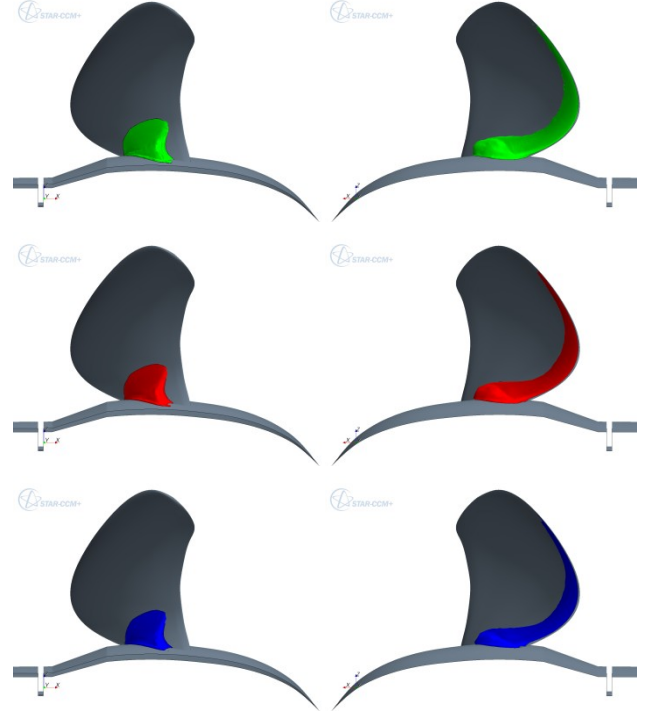


Figure 39: RANS Cavitation on the suction (left) and pressure (right) side. $J_{tunnel} = 1.408$, $\sigma_N = 2.00$. Increasing (from top, 0.2, to bottom, 0.8) fraction of vapour.

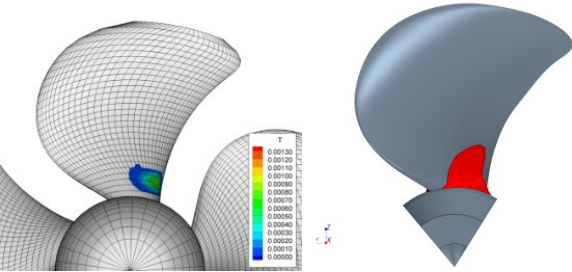


Figure 38: Cavitation on the suction side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.408$, $\sigma_N = 2.00$.

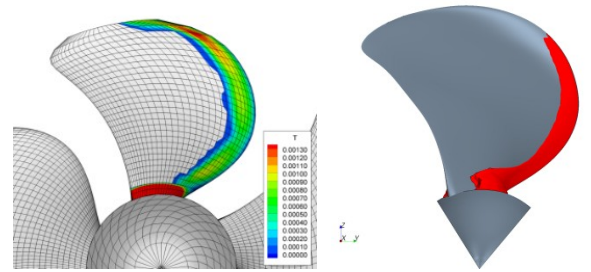


Figure 40: Cavitation on the suction side. Panel (left) vs. RANS (right). $J_{tunnel} = 1.408$, $\sigma_N = 2.00$.

If at $r/R = 0.7$ (figure 31) the non cavitating pressure distributions computed by the two solvers are very close, (especially for what regard the magnitude of the peak) and the corresponding cavity extensions, evaluable from the constant pressure regions of figure 34, are identical, moving towards the blade tip potential pressure

recovery region. Local pressure distribution is very close to the cavitation index but, at the same time, the pressure gradient is adverse to the cavity growth. Moreover the local discretization of the blade surface is such that the length of the local bubble is lower than the chordwise size of panels. As a consequence only one point on the

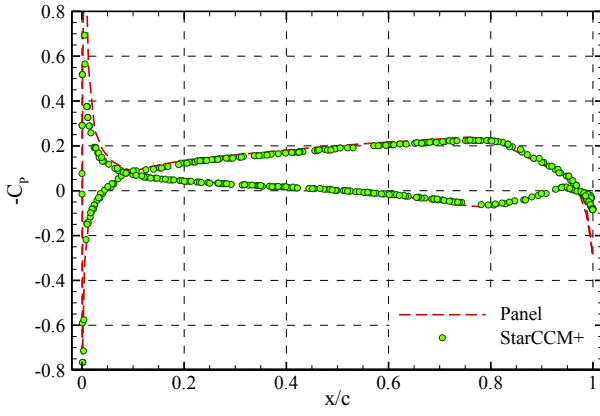


Figure 41: Non cavitating pressure distribution. $r/R = 0.7$
 $J_{\text{tunnel}} = 1.408$.

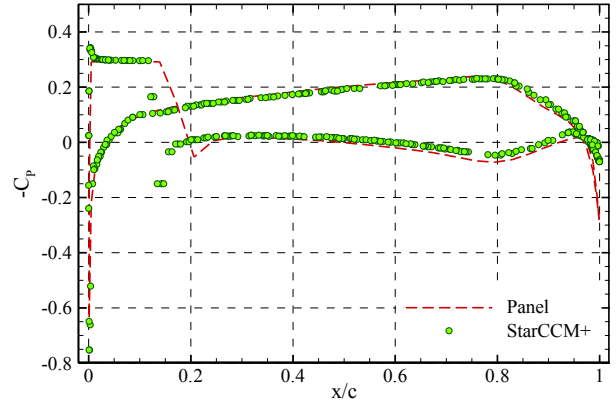


Figure 44: Cavitating pressure distribution. $r/R = 0.7$
 $J_{\text{tunnel}} = 1.408$, $\sigma_N = 2.00$.

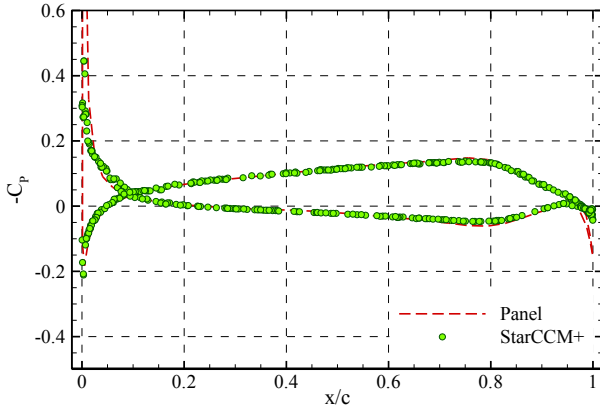


Figure 42: Non cavitating pressure distribution. $r/R = 0.9$
 $J_{\text{tunnel}} = 1.408$.

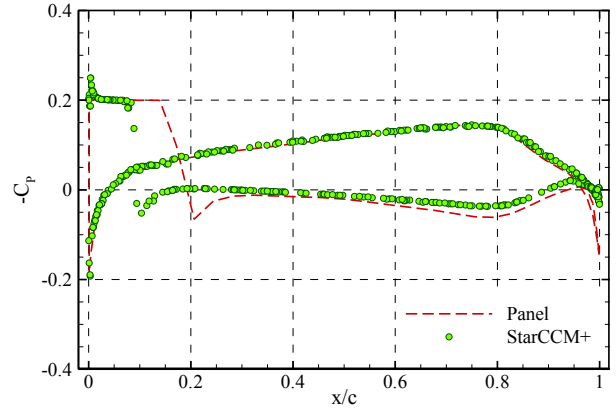


Figure 45: Cavitating pressure distribution. $r/R = 0.9$
 $J_{\text{tunnel}} = 1.408$, $\sigma_N = 2.00$.

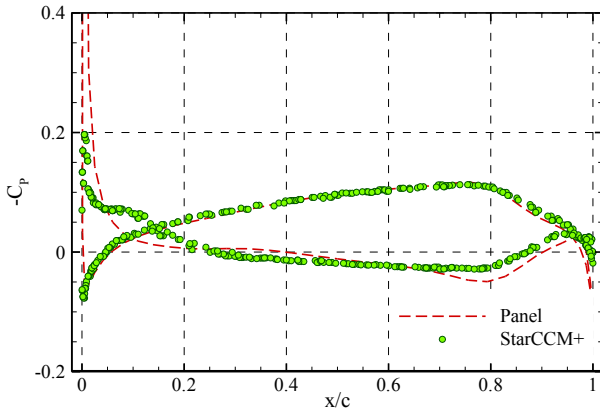


Figure 43: Non cavitating pressure distribution. $r/R = 0.95$
 $J_{\text{tunnel}} = 1.408$.

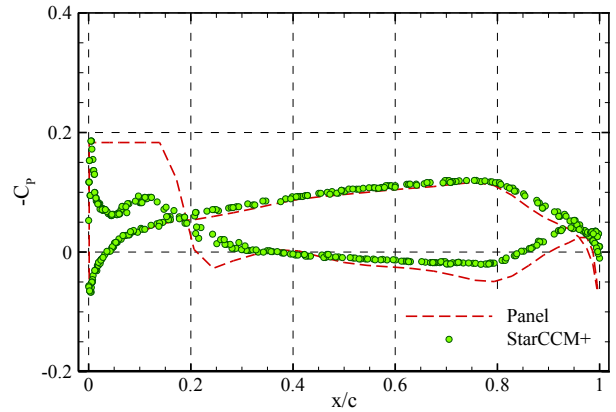


Figure 46: Cavitating pressure distribution. $r/R = 0.95$
 $J_{\text{tunnel}} = 1.408$, $\sigma_N = 2.00$.

pressure diagrams falls into the cavitating region, local cavity thickness is not computed (as in figure 28) and the presence of midchord bubble is recognizable only through pressure distribution and not through cavity thickness on the blade. Figures from 37 to 46, finally, compare cavity bubble extensions and pressure distributions for the flow conditions of case 2.3.3. At this higher value (1.408) of advance coefficient the risk of negative angle of attack is very high. In fact, as predicted similarly by the two

methodologies, cavitation is mainly of sheet type at pressure side blade leading edge. As for the previous computations Panel Method predictions tend to slightly overestimate (with respect to RANS) cavity extension, especially in the radial direction. Non cavitating and cavitating pressure distributions confirm this peculiarity, with inviscid pressure peaks (and resulting risk of cavitation) at leading edge strongly overestimated with respect to viscous computations. Back midchord root

cavitation is also similarly predicted by the two approaches. In terms of propeller performances differences between Panel Method and RANS solver are more important, not specifically for the absolute values (that, really, are close), but for their different behavior as they are calculated by the two solvers in cavitating conditions. In particular for case 2.3.1 (higher propeller load), while the RANS solution predicts a slightly lower value of thrust coefficient with respect to the non cavitating one, Panel Method estimates a propeller delivered thrust slightly higher than the non cavitating reference value. In the other two cases, instead, the tendency is the same for both the solvers. Even if for case 2.3.2 the RANS cavitating thrust coefficient is sensibly lower than that in non cavitating conditions (while the difference, noncavitating/cavitating, computed by the Panel Method is not so strong), both the solvers estimate values of thrust in cavitating conditions lower than the corresponding thrust in non cavitating flow. For case 2.3.3 even the reduction, with respect to the reference non cavitating condition, computed by the two methodologies is very similar, with relative differences around 5%. A reason for these differences, at least for what regards the developed panel method, for which a complete control on the solution, not allowed in a commercial solver, is possible, can be identified on the pressure recovery law adopted at the bubble trailing edge. Numerically, and depending on the flow conditions (angle of attack, non cavitating pressure distribution), the potential solution can be characterized by a constant vapor pressure region suddenly matched with the non cavitating pressure distribution downstream the bubble trailing edge. This sudden change in pressure is unrealistic: re-entrant jets and local condensation (that cannot be deal with a potential approach for with cavity bubble is a streamsurface of the flow) locally change the pressure distribution (and, as a consequence, forces) on the profile. Pressure recovery laws are approximate ways to model these effects, but they are mainly based on empirical factors, whose choice could alter the solution, especially in the transition regime between non cavitating and fully developed cavity bubble, in correspondence of which higher values of thrust and torque, with respect to the non cavitating regime, can be computed. Also the approximation in the calculation of bubble close to the tip (figure 26) can explain some of the differences.

4 CONCLUSIONS

Potential based panel methods and RANS viscous solvers have been applied for the computations of open water propeller performances, downstream induced velocities and cavity extension. Satisfactory results have been obtained with both the methodologies. The systematic comparison of results highlight the peculiarities of such different approaches. In open water conditions, at least for thrust prediction, all the employed numerical solvers predict very close results. Torque values are a bit more scattered, but except for OpenFOAM and Panel Method at lower advance coefficient, all the approaches seem

reliable. Non cavitating pressure distribution are in well agreement, with minor differences at blade leading and trailing edge. Also in cavitating conditions viscous and potential computations are very close. Except tip vortex cavitation both the solvers predict comparable cavity bubble extension, not only for the more typical leading edge back cavitation, but also for the face and midchord root cases. Cavitating performances predictions seem reliable.

All the methodologies can be considered, in conclusion, appropriate solvers for dedicated problems. Panel Method results in a fast and overall accurate solver, reliable in the design and optimization stage. RANS solvers, instead, due to not yet common required computation resources, can be applied for the final design analysis, in order to better understand flow phenomena (tip/hub vortex cavitation, for instance) responsible of noise and vibrations. OpenFOAM, on the other hand, represents a valid free alternative in the framework of viscous flow solvers. An accurate preliminary validation of the code and a more extensive training is still required.

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